Chemical Bonding

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CHEMICAL BONDING

1. TYPES
Ionic: metal + nonmetal
Covalent: nonmetals

2. LEWIS STRUCTURES
Count valence, arrange, connect,
add lone pairs, check octets

Exceptions:

P, S, Cl (expanded)

3. VSEPR

LP-LP > LP-BP > BP-BP

4. HYBRIDIZATION
sp, sp , sp , sp d, sp d

5. MO THEORY
Bonding < Antibonding
Bond order = (b-a)/2

6. POLARITY

0.4-1.7 polar
>1.7 ionic

Symmetric -> nonpolar
Asymmetric -> polar

7. IMF
London < Dipole < H-bond < Ion-dipole